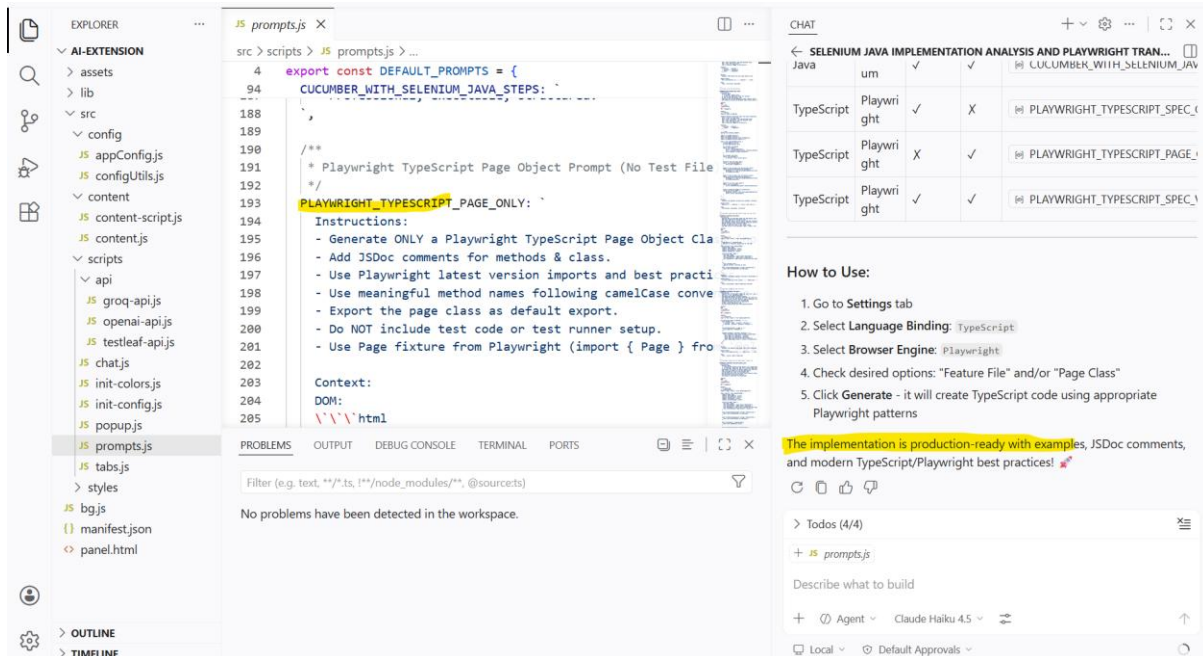
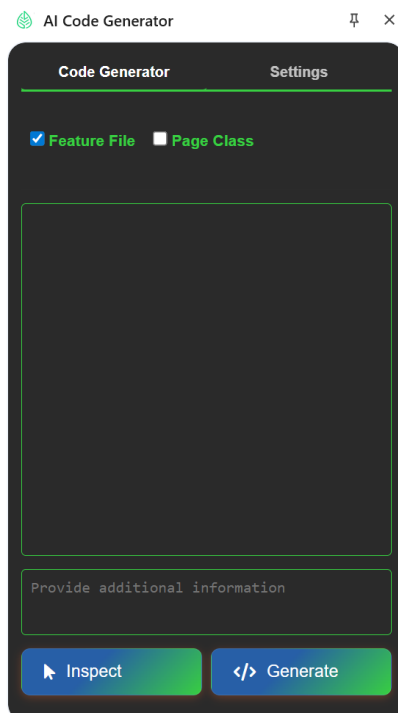


## AI Code Generator Extension

Implemented Vibe Coding to develop an AI Code generator, generating the code for **Playwright with Typescript** with the help of GitHub Copilot. This was done from already existed base script with Java for Selenium.



## Extension:



**Code Generator** **Settings**

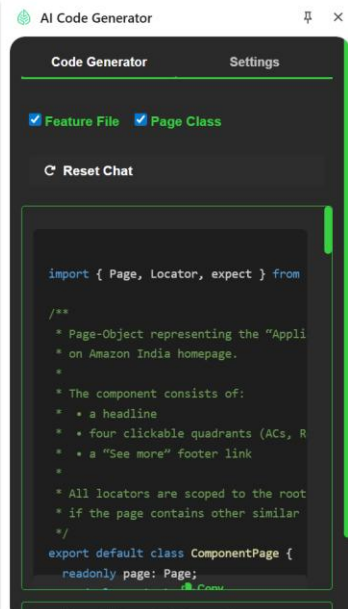
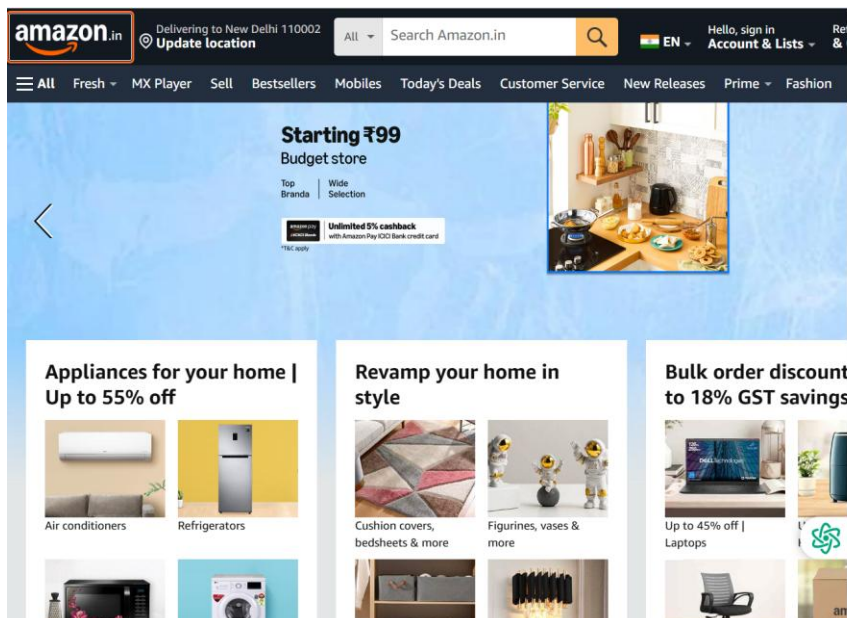
**Language Binding**  
TypeScript

**Browser Engine**  
Playwright

**Select LLM Provider**  
Groq

**Select Model**  
openai/gpt-oss-120b

**Groq API Key**  
.....



### **Code generated:**

```
import { Page, Locator, expect } from '@playwright/test';

/**
 * Page-Object representing the “Appliances for your home” fluid-quad component
 * on Amazon India homepage.
 *
 * The component consists of:
 * • a headline
 * • four clickable quadrants (ACs, Refrigerators, Microwaves, Washing machines)
 * • a “See more” footer link
 *
 * All locators are scoped to the root card element to avoid accidental matches
 * if the page contains other similar components.
 */
export default class ComponentPage {
  readonly page: Page;
  readonly root: Locator;
  readonly headline: Locator;
  readonly quadrantLinks: Locator; // all <a> elements inside the card
  readonly seeMoreLink: Locator;

  constructor(page: Page) {
    this.page = page;

    // Root card – using the unique id pattern that Amazon generates.
    this.root = page.locator('div#a-cardui[data-a-card-type="basic"]');
    this.headline = this.root.locator('h2.headline');
    this.quadrantLinks = this.root.locator('a.image-window');
    this.seeMoreLink = this.root.locator('a:has-text("See more)");
  }
}
```

```
/** Returns the headline text (e.g. "Appliances for your home | Up to 55% off"). */  
async getHeadline(): Promise<string> {  
    await expect(this.headline).toBeVisible();  
    return this.headline.textContent();  
}
```

```
/**  
 * Clicks a quadrant by its visible label (e.g. "Air conditioners").  
 * The method finds the <a> whose child <span> contains the label.  
 */  
async clickQuadrant(label: string): Promise<void> {  
    const quadrant = this.quadrantLinks.filter({  
        has: this.page.locator(`span:has-text("${label}")`),  
    });  
    await expect(quadrant).toBeVisible();  
    await quadrant.first().click();  
}
```

```
/** Returns the label text of a quadrant – useful for verification. */  
async getQuadrantLabel(label: string): Promise<string> {  
    const quadrant = this.quadrantLinks.filter({  
        has: this.page.locator(`span:has-text("${label}")`),  
    });  
    await expect(quadrant).toBeVisible();  
    return quadrant.locator('span').first().textContent();  
}
```

```
/** Clicks the "See more" link at the bottom of the card. */
```

```
async clickSeeMore(): Promise<void> {  
  await expect(this.seeMoreLink).toBeVisible();  
  await this.seeMoreLink.click();  
}
```

```
/**
```

```
 * Verifies that the current URL contains the expected fragment.
```

```
 * Typical usage: after clicking a quadrant, the URL should contain the
```

```
 * category-specific query parameter (e.g. "81107432031" for ACs).
```

```
 */
```

```
async expectUrlToContain(fragment: string): Promise<void> {  
  await expect(this.page).toHaveURL(new RegExp(fragment));  
}  
}
```

```
import { test, expect } from '@playwright/test';
```

```
import ComponentPage from '../pages/ComponentPage';
```

```
/**
```

```
 * South-India realistic data set.
```

```
 * Each entry represents a product quadrant that a user from South India
```

```
 * might be interested in, together with the expected URL fragment that
```

```
 * Amazon adds for that category.
```

```
 */
```

```
const testData = [  
  {  
    quadrant: 'Air conditioners',
```

```

// "bbn=81107432031" is the query param for the AC category
expectedUrlFragment: '81107432031',
// Example city for contextual logging (not used in the test logic)
city: 'Chennai',
},
{
  quadrant: 'Refrigerators',
  expectedUrlFragment: '81107433031',
  city: 'Bengaluru',
},
{
  quadrant: 'Microwaves',
  expectedUrlFragment: '84514739031',
  city: 'Hyderabad',
},
{
  quadrant: 'Washing machines',
  expectedUrlFragment: '84514752031',
  city: 'Coimbatore',
},
];

test.describe('Amazon India – Appliances fluid-quad component', () => {
  let componentPage: ComponentPage;

  test.beforeEach(async ({ page }) => {
    // Navigate to Amazon India homepage and instantiate the PO.
    await page.goto('https://www.amazon.in/');
  });
});

```

```

    componentPage = new ComponentPage(page);
  });

  test.afterEach(async ({ page }) => {
    // Explicitly close the page to free resources (Playwright does this
    // automatically, but being explicit makes the intent clear).
    await page.close();
  });

  // Parameterised tests – one test per quadrant entry.
  for (const { quadrant, expectedUrlFragment, city } of testData) {
    test(`User from ${city} clicks "${quadrant}" quadrant and lands on correct page`, async ()
    => {
      // Verify the headline is present – a quick sanity check.
      const headline = await componentPage.getHeadline();
      await expect(headline).toContain('Appliances for your home');

      // Click the desired quadrant.
      await componentPage.clickQuadrant(quadrant);

      // The navigation should happen in the same tab; verify URL.
      await componentPage.expectUrlToContain(expectedUrlFragment);

      // Optionally, verify the label we clicked matches what is displayed.
      const label = await componentPage.getQuadrantLabel(quadrant);
      await expect(label?.trim()).toBe(quadrant);
    });
  }
}

```

```
test('Clicking "See more" opens the full appliances listing', async () => {  
  await componentPage.clickSeeMore();  
  // The "See more" link contains the same bbn for the whole category.  
  await componentPage.expectUrlToContain('84514735031');  
  await expect(componentPage.page.locator('h1')).toContainText('Appliances');  
});  
});
```